Linear Transformations

Recall from a previous class:

- A set of vectors $\{\vec{v}_1, \ldots, \vec{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_p\vec{v}_p = \vec{0}$$

  has only the trivial solution. The set is called **linearly dependent** if the equation has non-trivial solutions and the equation is called the **dependence relation**.

- Let $A = [\vec{a}_1 \ \ldots \ \vec{a}_n]$, then $\{\vec{a}_1, \ldots, \vec{a}_n\}$ are linearly independent if and only if $A\vec{x} = \vec{0}$ has only the trivial solution.

- **Theorem 7:** A set $S = \{\vec{v}_1, \ldots, \vec{v}_p\}$ of two or more vectors is linearly dependent if and only if one of the vectors of S is a linearly combination of the others.

  Moreover if $\vec{v}_1 \neq \vec{0}$, then some $\vec{v}_j$ for $j > 1$ is a linear combination of $\vec{v}_1, \ldots, \vec{v}_{j-1}$.

- **Theorem 8:** Any set $\{\vec{v}_1, \ldots, \vec{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

Define $T : \mathbb{R}^n \to \mathbb{R}^m$ by $T(\vec{x}) = A\vec{x}$ where A is an m by n matrix. Recall we call $\mathbb{R}^n$ the domain and $\mathbb{R}^m$ the codmain. The set of outputs, $\{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\}$, is the image or the range of T.

Definition 1. *T is* **linear** *if*

1. *$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v}$ in $\mathbb{R}^n$,*
2. *$T(c\vec{u}) = cT(\vec{u})$ for all $\vec{u}$ in $\mathbb{R}^n$ and c in $\mathbb{R}$.*

1. Properties of Linear Transformations.

(a) Prove if T is linear, then $T(\vec{0}) = \vec{0}$.

(b) $T(c\vec{u} + d\vec{v}) = cT(\vec{u}) + dT(\vec{v})$.

2. Let $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \\ 2 & -2 \end{bmatrix}$ and let $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$.

(a) Is $\vec{b}$ in the range of T?

(b) What is the range of T?

(c) Is T surjective?

(d) Is the transformation T injective?

3. Let $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} 1 & 0 & -3 \\ -3 & 1 & 6 \\ 2 & -2 & -1 \end{bmatrix}$ and let $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$. Is $\vec{b}$ in the range of T?

4. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear and $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ be a linearly dependent set. Is $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$ linearly dependent?

5. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear and $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ be a linearly **in**dependent set. Is $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$ linearly **in**dependent?